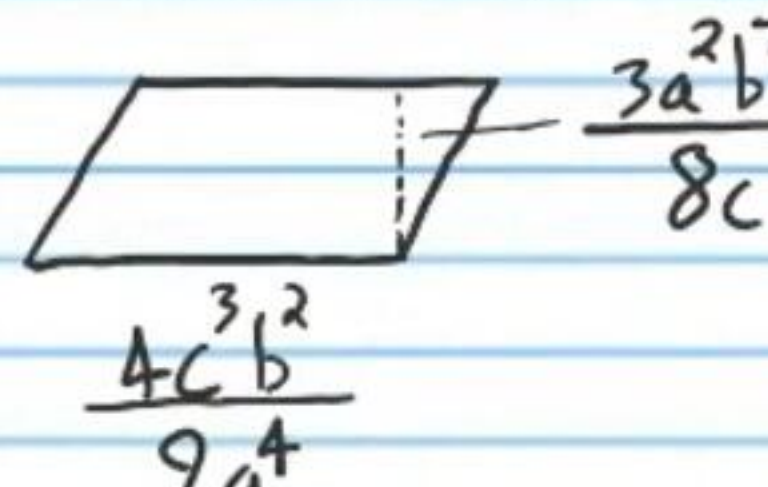


Pre-Algebra Practice Final Test-Version I

Instructions

- As always, write a title, name, date, and course.
- Number each problem, circle the number, and show all work.
- Draw a rectangle around your answers.
- Rewrite the equation whenever you perform an operation on both sides to preserve the result of the previous step.

① Find the area of the parallelogram below.



$$A = bh = \frac{4c^3b^2}{9a^4} \left(\frac{3a^2b^7}{8c} \right)$$

$$= \frac{4 \cdot 3 c^3 b^2 a^2}{9 \cdot 8 a^4 c}$$

$$= \frac{4 \cdot 3 c^2 b^9}{3 \cdot 3 \cdot 4 \cdot 2 a^2} = \boxed{\frac{c^2 b^9}{6a^2}}$$

Simplify the following expressions.

② $(-6.1x^2z^5y)(-0.87x^4y^3z^6) = \boxed{5.307x^6z^{11}y^4}$

③ $-\frac{12x^5}{5y^3} \div 8x^7y^5 = -\frac{12x^5}{5y^3} \div \frac{8x^7y^5}{1}$

$$= -\frac{12x^5}{5y^3} \left(\frac{1}{8x^7y^5} \right)$$

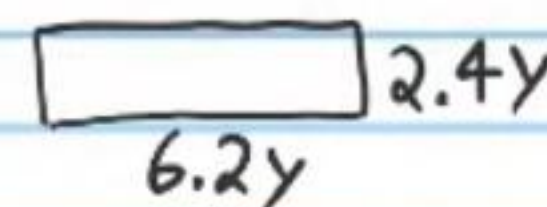
$$= -\frac{12x^5}{5 \cdot 8 x^7 y^8} = \frac{-3}{10x^2y^8} = \boxed{\frac{-3}{10x^2y^8}}$$

④ a) $\frac{81a^{10}b^9}{-9a^5b^{11}} = \boxed{-\frac{9a^5}{b^2}}$ b) $\frac{24x^6y^3}{30xy^4} = \boxed{\frac{4x^5}{5y}}$

⑤ $\frac{9abc}{10} \div \left(-\frac{ab}{4c} \right) = \frac{9abc}{10} \left(-\frac{4c}{ab} \right) = \frac{-9 \cdot 4 abc^2}{10ab}$

$$= \frac{-9 \cdot 2 \cdot 2 c^2}{2 \cdot 5} = \boxed{-\frac{18c^2}{5}}$$

⑥ a) Find the perimeter of the rectangle below.



$$P = 6.2y + 2.4y + 2.4y + 6.2y$$

$$P = \boxed{17.2y}$$

b) Find the volume of a sphere with radius $3x^2y$.

$$V = \frac{4\pi r^3}{3} = \frac{4\pi (3x^2y)^3}{3}$$

$$= \frac{4\pi (3x^2y)(3x^2y)(3x^2y)}{3}$$

$$= \boxed{36\pi x^6y^3}$$

⑦ Solve the equation below. Then check your answer.

a) $-x + 2x - 3 = 3x + 1 - 4x$

$$x - 3 = -x + 1$$

$$x + x - 3 = -x + 1 + x$$

$$2x - 3 = 1$$

$$3 + 2x - 3 = 1 + 3$$

$$2x = 4$$

$$\frac{2x}{2} = \frac{4}{2}$$

$$\boxed{x = 2}$$

b) Check

$$-2 + 2(2) - 3 = 3(2) + 1 - 4(2)$$

$$-2 + 4 - 3 = 6 + 1 - 8$$

$$\boxed{-1 = -1}$$

Yes

Solve.

$$\textcircled{8} -3(2y+3)=5-(y+2)-4y \quad \textcircled{9} -\frac{z}{5}+\frac{3z}{10}+1=4+(-8)$$

$$-6y-9=5-y-2-4y$$

$$-6y-9=3-5y$$

$$\boxed{-12=y}$$

$$-\frac{2z}{10}+\frac{3z}{10}+1=-4$$

$$\frac{z}{10}+1=-4$$

$$-1+\frac{z}{10}+1=-4-1$$

$$\frac{z}{10}=-5$$

$$(10)\frac{z}{10}=-5(10)$$

$$\boxed{z=-50}$$

Evaluate.

$$\textcircled{10} 4x^3+7x-x^2+1, x=-2 \quad \textcircled{11} -7z-z^2+8, z=3.$$

$$4(-2)^3+7(-2)-(-2)^2+1$$

$$4(-8)+7(-2)-4+1$$

$$-32-14-4+1$$

$$\boxed{-49}$$

$$-7(3)-3^2+8$$

$$-7(3)-9+8$$

$$-21-9+8$$

$$\boxed{-22}$$

$$\textcircled{12} \frac{y^2-1}{-3y}-2(3-2y)^2, y=1 \quad \textcircled{13} 5(x+3)^2-4x+x^3, x=-1$$

$$\frac{1^2-1}{-3(1)}-2(3-2(1))^2$$

$$\frac{1-1}{-3}-2(3-2)^2$$

$$\frac{0}{-3}-2(1)^2$$

$$0-2(1)$$

$$0-2$$

$$\boxed{-2}$$

$$5(-1+3)^2-4(-1)+(-1)^3$$

$$5(2)^2+4-1$$

$$5(4)+4-1$$

$$20+4-1$$

$$\boxed{23}$$

$$\textcircled{14} \text{ Find } D \text{ if } D=b^2-4ac, b=2, a=-2, \text{ and } c=1.$$

$$D=b^2-4ac=2^2-4(-2)(1)$$

$$=4-4(-2)(1)$$

$$=4-4(-2)$$

$$=4+8=\boxed{12}$$

$$\textcircled{15} \text{ a) Evaluate your answer for problem \#1 if } a=2, b=1, \text{ and } c=3.$$

$$\frac{c^2b^9}{6a^2}=\frac{3^2(1)^9}{6(2)^2}=\frac{9(1)}{6(4)}$$

$$=\frac{9}{24}=\boxed{\frac{3}{8}}$$

$$\text{b) Evaluate your answer for problem \#6a if } y=-2.$$

$$P=17.2y=17.2(-2)$$

$$=\boxed{-34.4}$$

Solve. Round to the nearest thousandth if necessary.

$$\begin{aligned} \textcircled{16} \quad 3 - 2.7x + 5x &= 1 - 3.06x \\ 3 + 2.3x &= 1 - 3.06x \\ 3.06x + 3 + 2.3x &= 1 - 3.06x + 3.06x \\ 3 + 5.36x &= 1 \\ -3 + 3 + 5.36x &= 1 - 3 \\ 5.36x &= -2 \\ \frac{5.36x}{5.36} &= \frac{-2}{5.36} \\ x &\approx -0.373134 \\ \boxed{x \approx -0.373} \end{aligned}$$

$$\begin{aligned} \textcircled{18} \quad 2(3 - 2z) - 4(-2z + 1) &= z - 3(3z - 2) + 8z \\ 6 - 4z + 8z - 4 &= z - 9z + 6 + 8z \\ 4z + 2 &= 6 \\ 4z &= 4 \\ \boxed{z = 1} \end{aligned}$$

$$\begin{aligned} \textcircled{17} \quad -3y - 3y + 7 &= -7y + 1 \\ -6y + 7 &= -7y + 1 \\ 7y - 6y + 7 &= -7y + 1 + 7y \\ y + 7 &= 1 \\ -7 + y + 7 &= 1 - 7 \\ \boxed{y = -6} \end{aligned}$$

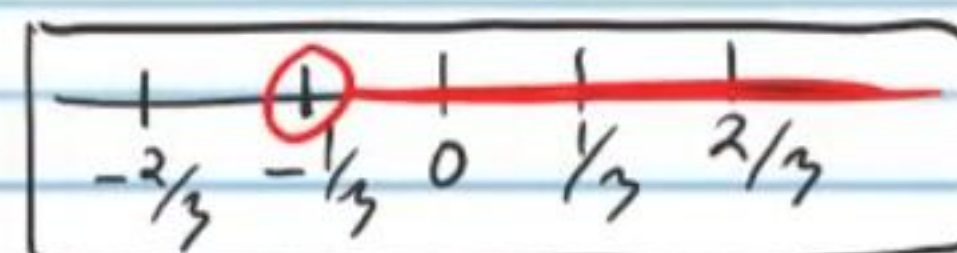
Find the algebraic solutions and the graphical solutions of the following inequalities.

$$\begin{aligned} \textcircled{19} \quad 2x &< 1 + 5x \\ -5x + 2x &< 1 + 5x - 5x \end{aligned}$$

$$-3x < 1$$

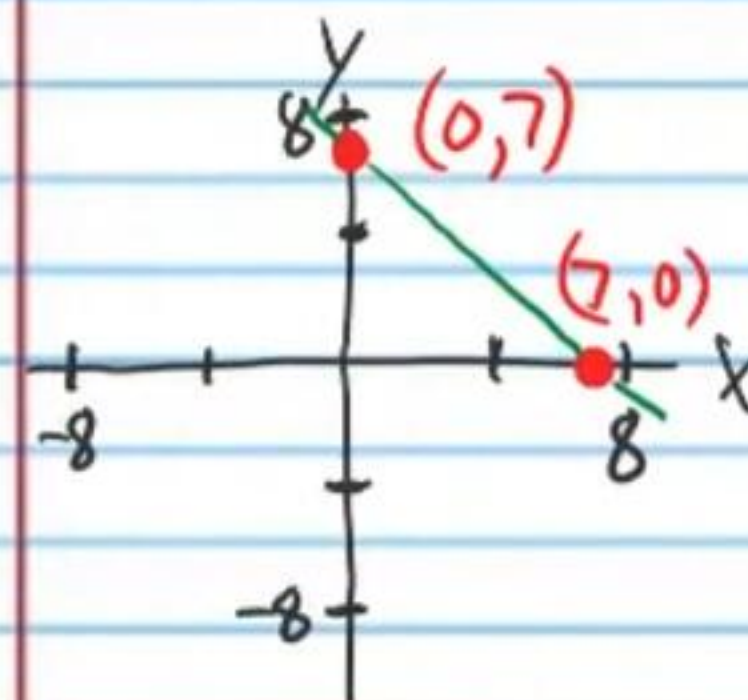
$$\frac{-3x}{-3} < \frac{1}{-3}$$

$$\boxed{x > -\frac{1}{3}}$$



Graph the following equation. Show all work and identify the points you use to graph the equation. Label your graph properly.

$$\textcircled{21} \quad 7 = x + y$$



$$\begin{array}{c|c} x & y \\ \hline 0 & 7 \\ 7 & 0 \end{array}$$

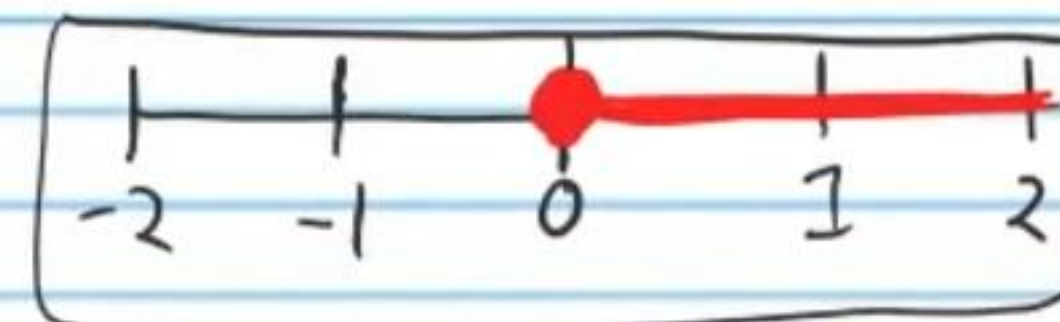
1 st Point	2 nd Point
$x = 0$	$y = 0$
$\Downarrow$	$\Downarrow$
$7 = 0 + y$	$7 = x + 0$
$7 = y$	$7 = x$
$(0, 7)$	$(7, 0)$

$$\textcircled{20} \quad -3(-5 - y) \geq y + 15$$

$$15 + 3y \geq y + 15$$

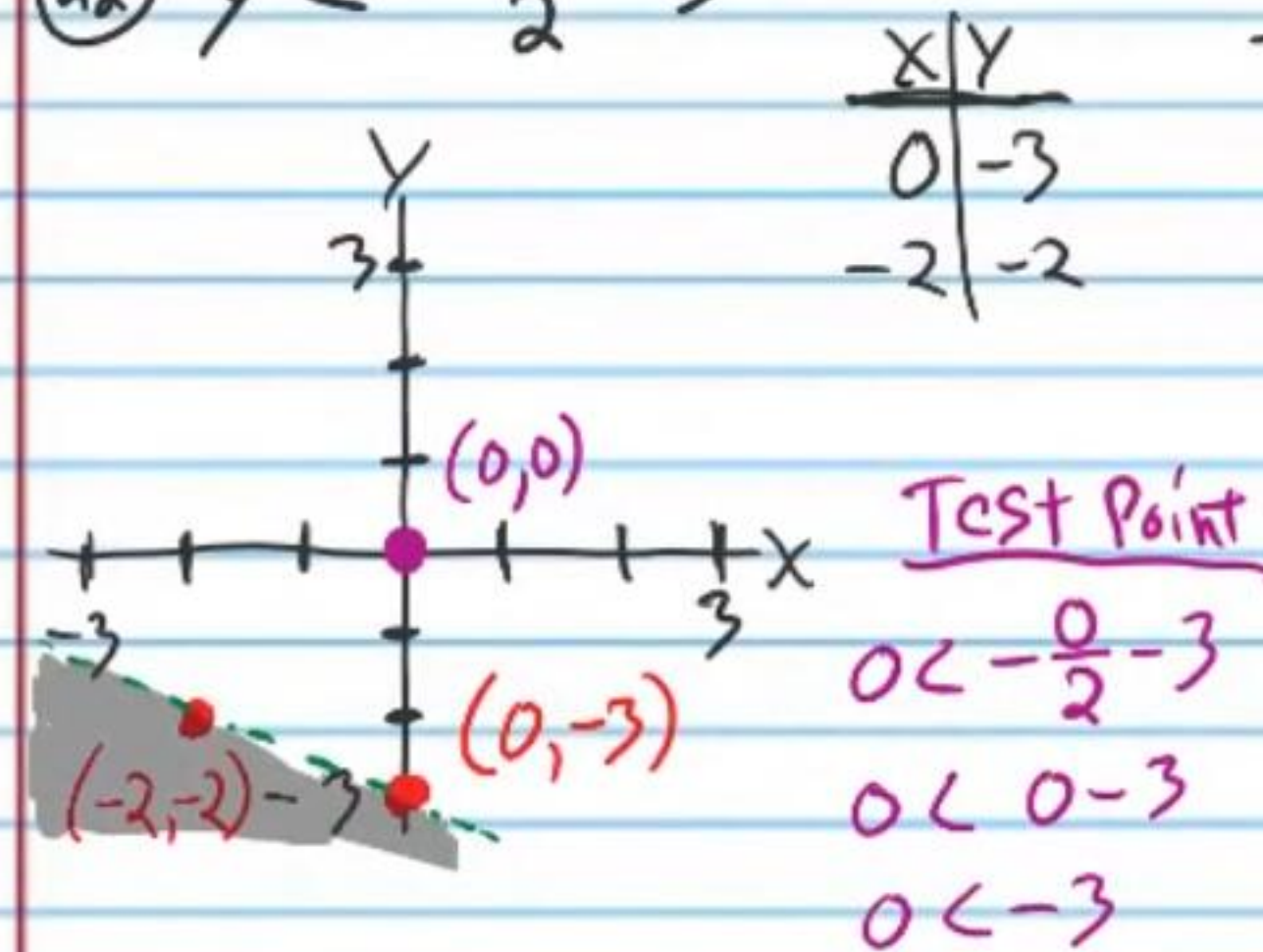
$$2y \geq 0$$

$$\boxed{y \geq 0}$$



Graph the following inequality. Show all work and identify the points you use to graph the inequality. Label the graph properly.

22) $y < -\frac{x}{2} - 3$



	1st Point	2nd Point
x/y	$x=0$	$x=-2$
	$y = -\frac{0}{2} - 3$	$y = -\frac{(-2)}{2} - 3$
	$y = 0 - 3$	$y = -(-1) - 3$
	$y = -3$	$y = 1 - 3$
	$(0, -3)$	$y = -2$
		$(-2, -2)$

Test Point
 $0 < -\frac{0}{2} - 3$
 $0 < 0 - 3$
 $0 < -3$

23) When a company raised its prices, net monthly profit decreased by 11.3%. Profit for this latest month was 18 million dollars. What was the monthly profit before the decrease? Round to the nearest hundredth of a million dollars.

$$N = F - Fr$$

$$18 = F - F(0.113)$$

$$18 = F - 0.113F$$

$$18 = 0.887F$$

$$\frac{18}{0.887} = \frac{0.887F}{0.887}$$

$$20.2931 \approx F$$

$$20.29 \approx F$$

20.29 million dollars

24) The sum of four consecutive even integers is 92. Identify these integers. Show the work that leads to your answer. You will receive no points for guessing with your calculator and writing the final answers

1st $\rightarrow x$ $x + (x+2) + (x+4) + (x+6) = 92$

2nd $\rightarrow x+2$ $x + x+2 + x+4 + x+6 = 92$

3rd $\rightarrow x+4$ $4x + 12 = 92$

4th $\rightarrow x+6$ $-12 + 4x + 12 = 92 - 12$

$$4x = 80$$

$$20, 22, 24, 26$$

$$\frac{4x}{4} = \frac{80}{4}$$

$$x = 20$$

25) You take out an \$8,000 loan and the bank charges 15% simple interest per month. Over time, your debt grows to \$15,200. Assuming that you didn't pay any of the money back, how much time passed since you received the money?

$$A = P + Prt$$

$$15,200 = 8,000 + 8,000(0.15)t$$

$$15,200 = 8,000 + 1,200t$$

$$-8,000 + 15,200 = 8,000 + 1,200t - 8,000$$

$$7,200 = 1,200t$$

$$\frac{7,200}{1,200} = \frac{1,200t}{1,200}$$

$$6 \text{ months} = t$$